What is Framework?

Framework is the guidelines or coding standards which we have to follow in order to automate test scripts.
Below are the framework which supports automation testing.

1) Data Driven Testing
2) BDD Cucumber
3) Keyword Driven Testing
4) Page Object Model (Not a framework, Design Pattern)
5) Page Factory (Extension of Page Object Model)
6) Hybrid Driven Testing Framework

Advantage :

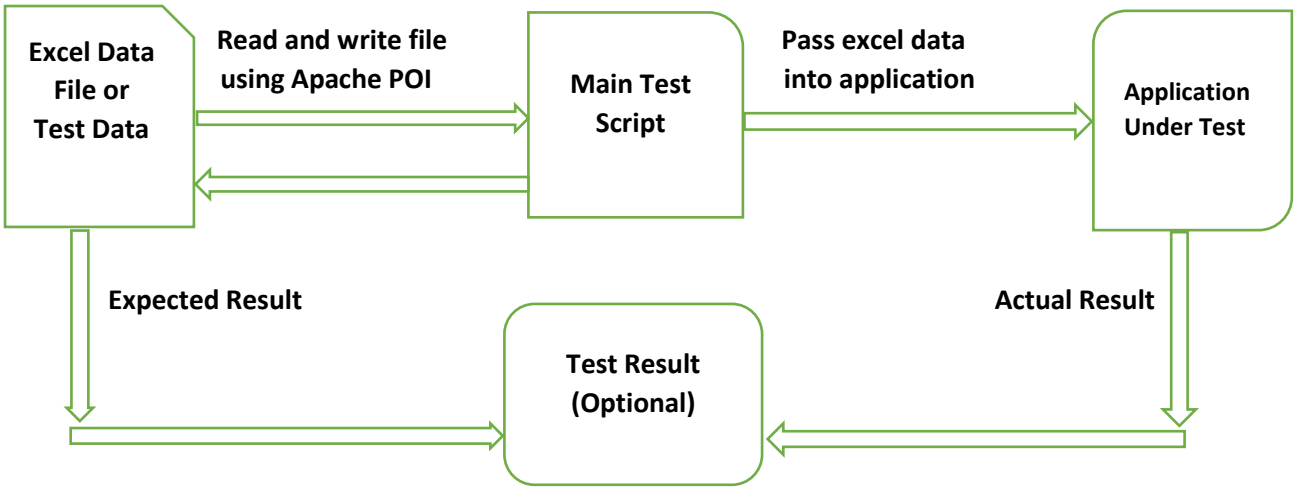
1) Code readability
2) Reuse utility file
3) Easy to understand

Data Driven Testing Framework :

Data driven testing is the automation testing framework where the test script reads or write test data in the excel file using Apache POI library and automate the test script in such a way that the excel data values are passed in order to interact with the front end of the application. It can be also used for validation.

Below are the libraries which is used for data driven testing.

1) poi 2) poi-ooxml 3) poi-ooxml-schemas 4) poi-ooxml-lite



Advantage :

1) Modify test data without changing the source code
2) Read the test data in excel file
3) Write the test data in excel file
4) Create utility file and easy to call the function

Disadvantage :

1) Require utility file
2) Debugging becomes difficult if test data goes wrong

BDD Cucumber Framework :

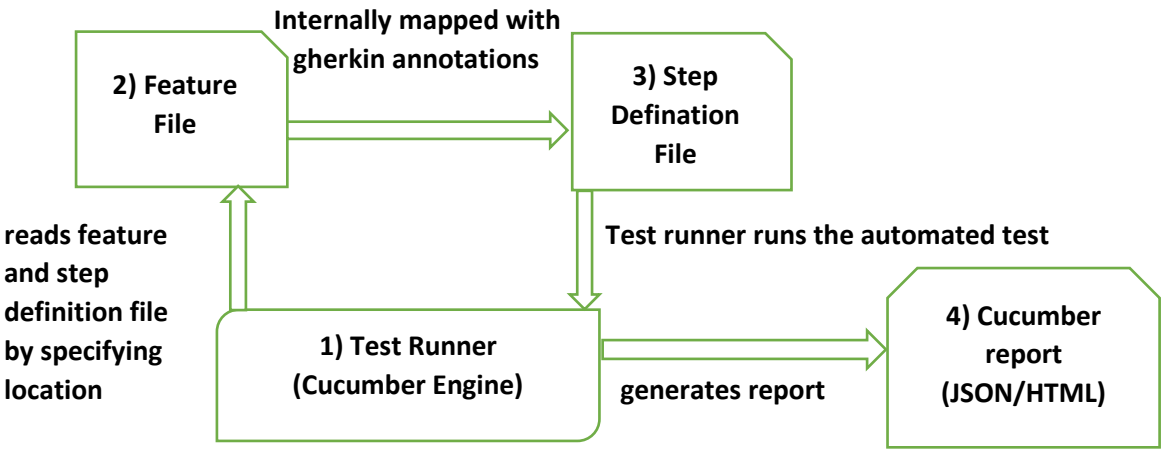
Behaviour Driven Development (BDD) is the software development approach which focus on testing the application behaviour.
Cucumber is one of the tool which comes under Behaviour Driven Development.
Cucumber is the BDD tool which runs the automated test script with the help of three components. They are 1) Feature File 2) Step Defination File 3) Test Runner. Below are the libraries which are used for BDD Cucumber java testing.
1) cucumber-java 2) cucumber-jvm 3) cucumber-junit 4) junit

Feature File : It is the file which helps to write business scenarios in a plain english language using gherkin keywords. Below are the gherkin keywords.
Feature : It describes high level functionality
Scenario : It describes test case
Scenario Outline : It is used for parameterization or data driven testing. It can be achieved with “Examples” keyword
Background : It is used when there are common steps or scenarios
Given : It defines pre conditions or necessary steps which are required to interact with application
When : It defines action or how we are going to interact with the application
Then : It defines expected result
Tags : It defines feature or functionality in the feature file

Step Defination File : Feature file is mapped to the step definition file with the help of gherkin annotations (given, when and then) Step definition file runs the automated test script. We can also internally call the page object file in step definition.

Test Runner : It is the file which is used for execution based on the location of feature file, location of step definition file, report and tags. It can be executed by Junit or TestNG. Below are the components which are specified in test runner file.

1) features : Defines location of feature file
2) glue : Defines location of step definition file
3) plugin : Generates report based on format like html or Json etc
4) monochrome : Makes the output clean
5) tags : Run feature file or test case based on the tags.



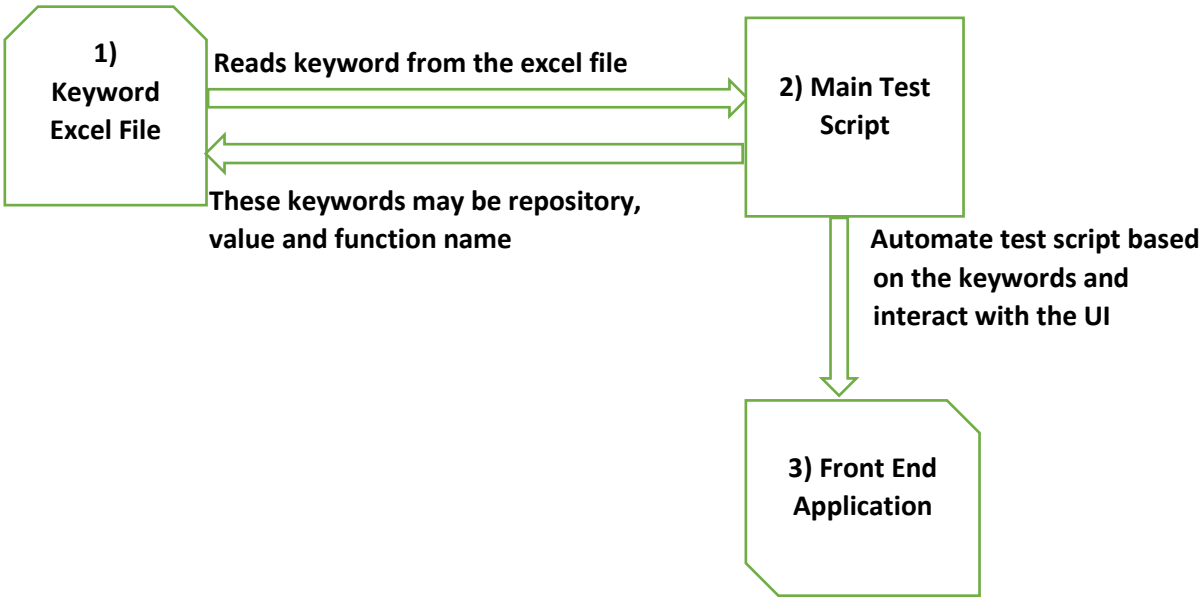
Advantage :

1) Easy to understand feature file. It is written in plain english language with gherkin keywords
2) It will be helpful for business analyst, stakeholders and non technical people to understand the functionality
3) Easy to understand test scripts
4) Proper documentation

Keyword Driven Testing Framework :

Keyword driven testing is the automation testing framework where it automates the test script based on the keywords present in the excel file. These keywords may be helpful to

1) **Maintain locators and locators value (Object Repository)**
2) **Maintain function name where you can call method to execute**
3) **Validate based on the functionality (eg. You can select or deselect common checkbox or radiobutton in the application based on the keyword excel file)**
4) **Pass values for the single web element (eg. You can pass particular value for the dropdown or form)**
5) **Maintain abstract methods from WebElement (Selenium WebDriver) interface. These abstract methods may be click(), sendKeys(), clear(), getTagName(), getAttribute(), getText(), , isDisplayed(), isEnabled(), isSelected() and findElements()**
6) **Maintain abstract methods from WebDriver (Selenium WebDriver) interface. These abstract methods may be getCurrentUrl(), getTitle(), close(), quit(), navigate() and findElements()**



Advantage :

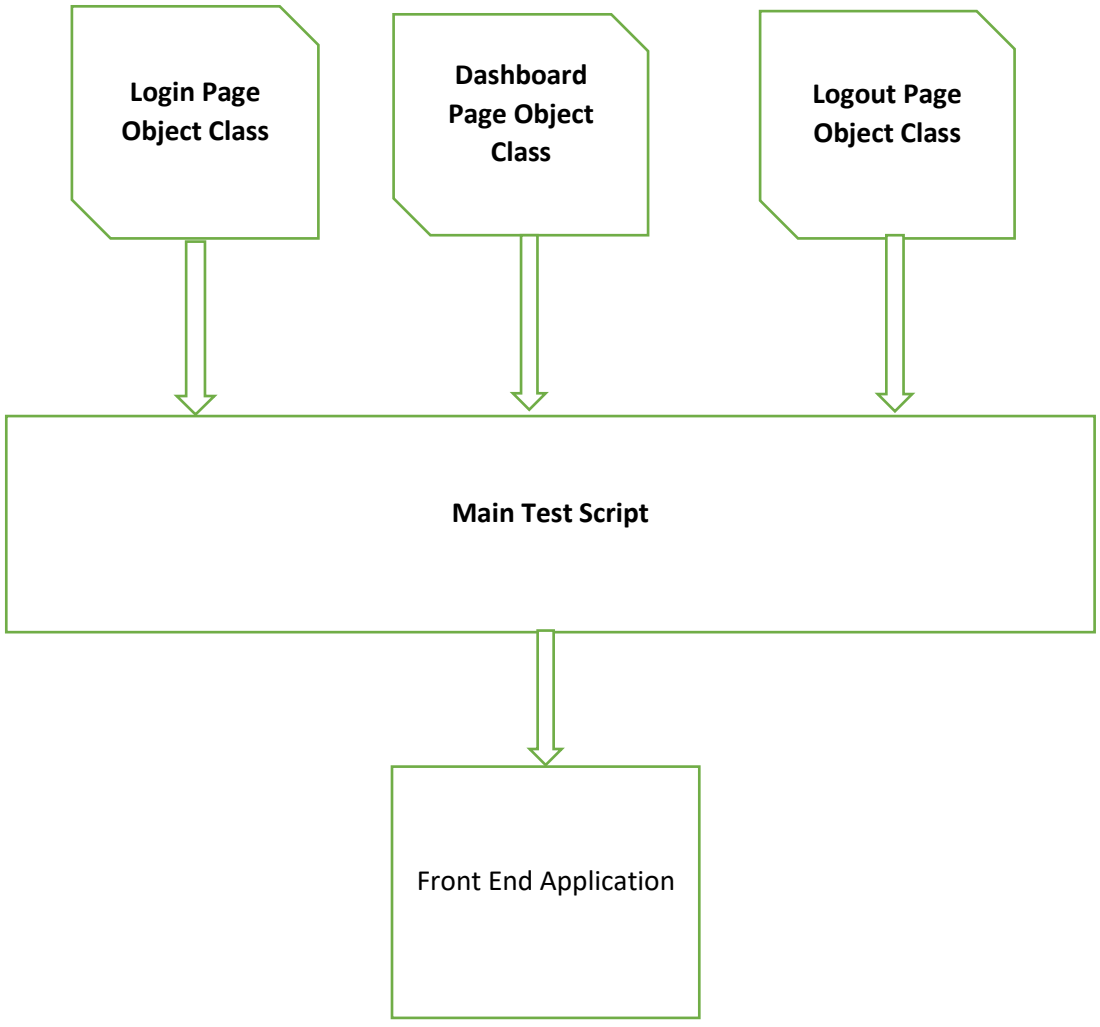
1) Faster Execution
2) Code Reusability
3) Separate files
4) Less maintenance

Disadvantage :

1) Debugging becomes difficult if test data in excel file are wrong

Page Object Model (POM) :

Page Object Model (POM) is the design pattern approach which helps to maintain the test script in separate files. Main test script and page object file are maintained in the separate files.
Main test script is maintained by Java or other programming language. All functions according to the test requirement and webpage are maintained in the page object file. It also consists of object repository. Here we need to ensure that the object of the interface (Selenium WebDriver or Playwright) should be initialized properly, Otherwise it will throw NullPointerException. Now this object contains abstract methods according to the library which helps us to interact with the front end of the application.
For execution, We have to first create object of the page object class (constructor). The moment we create object, Constructor will be automatically invoked and when we call the method or function with the help of object it will navigate to the function and execute.



Advantage :

1) Test script are in separate files
2) Code readability
3) Code reusability
4) Easy to understand
5) Faster execution

Page Factory :

Page Factory is the extension of page object model. It is the approach which helps to store and initialize the webelements at one place using Selenium WebDriver. Page Factory consists of @FindBy annotation and PageFactory.initElements(interface object) method. @FindBy annotation is used to store and find the webelements. Page Factory consists of initElements() method which helps to initialize the webelements. Thus, interface object present in the initElements() helps to invoke the method.

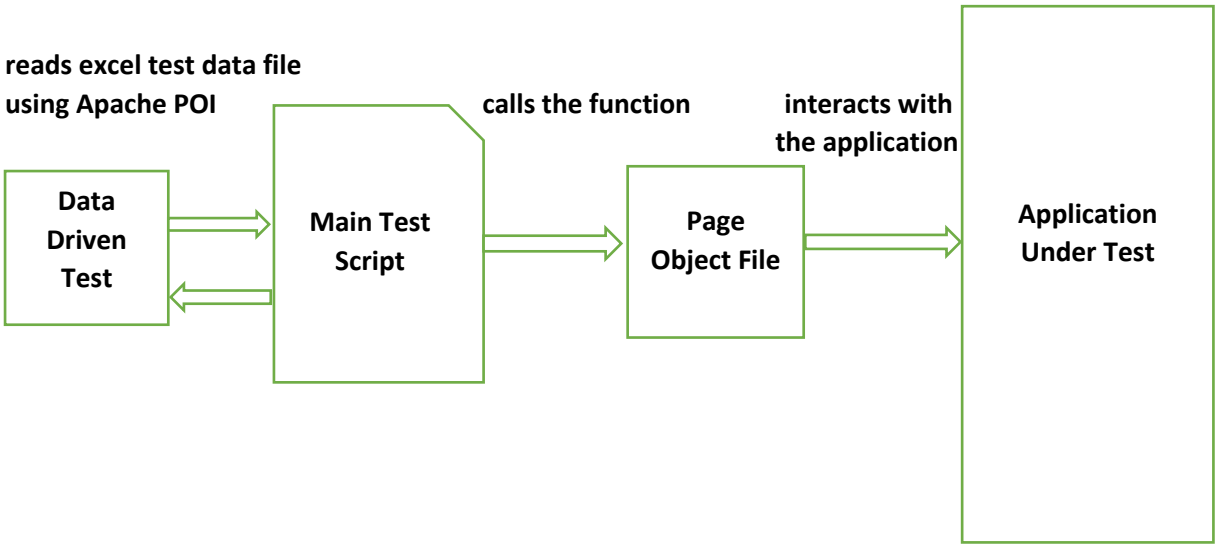
Advantage :

1) All webelements are stored at one place
2) Easy to understand

Hybrid Driven Testing Framework:

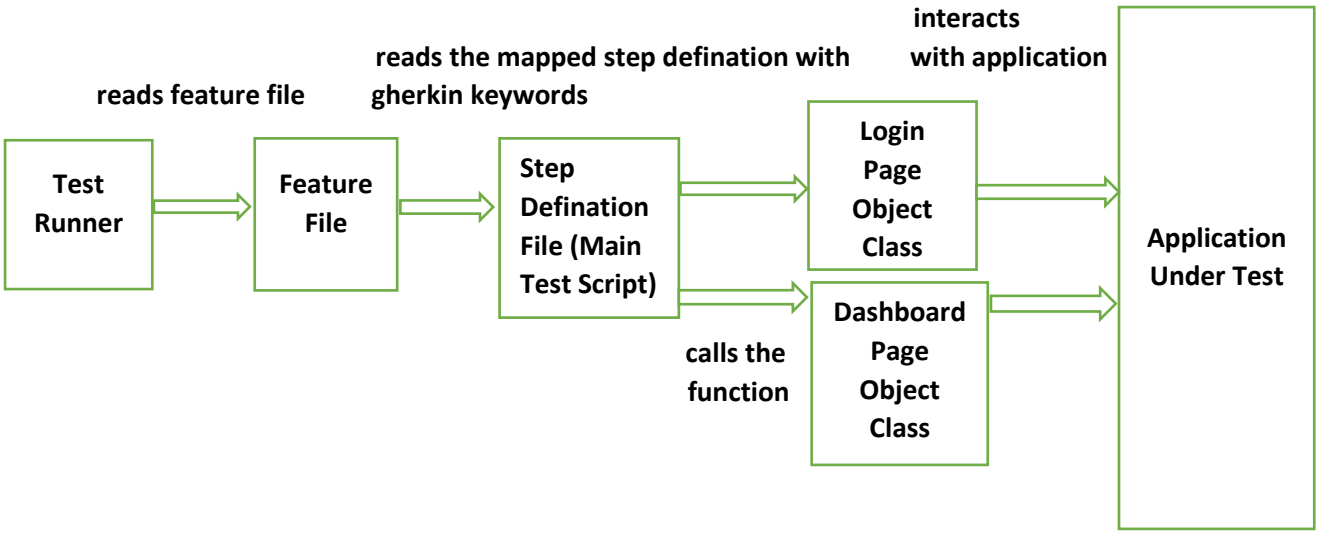
Hybrid Driven Testing framework is the combination of one or more framework. Below mentioned framework are commonly used

1) Data Driven Testing + Page Object Model



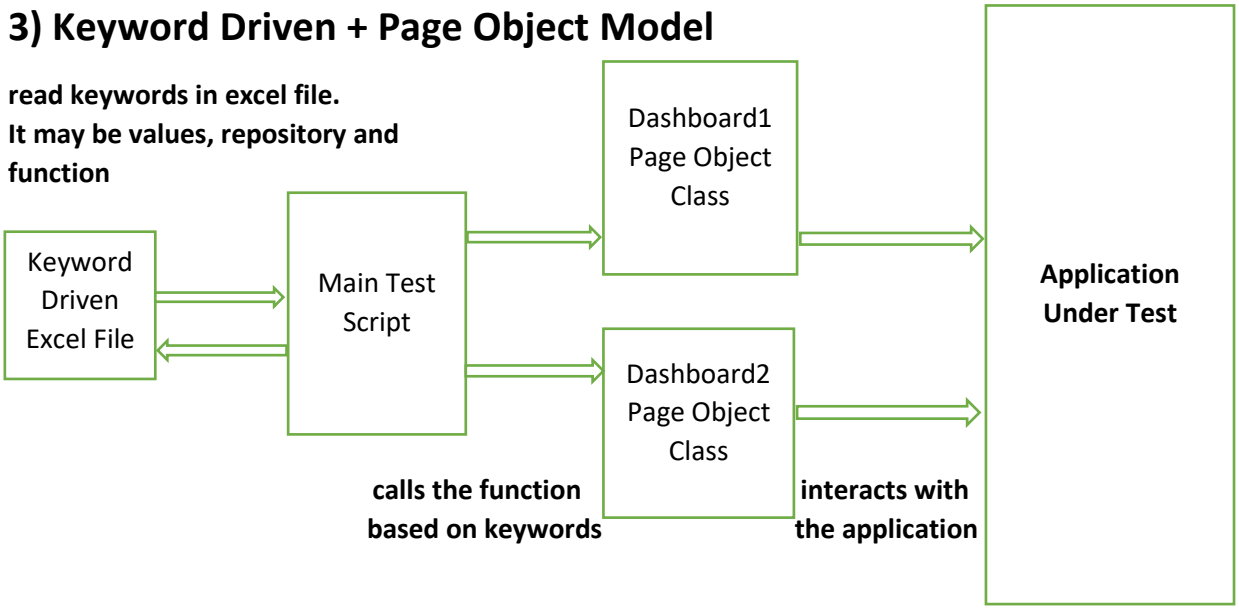
This combination helps to maintain test data in excel file. The main test script reads the excel test data using Apache POI library and calls the function which is present in the page object file in order to interact with the front end of the application.

2) BDD Cucumber + Page Object Model



This combination helps to maintain high level scenarios in the feature file, test script in the step defination file and functions in the page object file. Test runner reads the feature file. We can also parameterize test data in the feature file. Feature file reads the step defination file. Here feature file is mapped to the step defination file with the help of gherkin keywords. Test script in the step defination file calls the function in the page object class and interact with the front end of the application.

3) Keyword Driven + Page Object Model



This combination helps to maintain keywords in the excel file. These keywords may be action methods, locator, values of locator and function. Thus, Main test script file calls the function in the page object class based on the keywords in the excel file and interacts with the front end of the application.